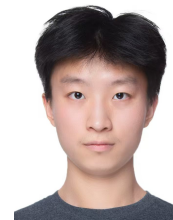


Haotian Liu

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Research Focus: Multimodal Agent Evaluation • Safety-Critical AI • Human-AI Interaction



EDUCATION

Xiamen University (XMU), Xiamen, China

Sep. 2023 – Jun. 2027 (expected)

B.Eng. in Software Engineering | Overall Average: 82.66/100

Relevant Coursework:

Machine Learning for Decision-Making and Optimization (96); Algorithms for Modern AI (94); Introduction to Artificial Intelligence (93); Interactive Design (90); The Practice of Computer Programming (92).

PUBLICATIONS

PilotBench: A Benchmark for General Aviation Agents with Safety Constraints.

IJCNN 2026, accepted. Co-first author.

arXiv:2604.08987

Long-Horizon Memory Governance for LLM Agents.

Under review at AAAI 2027. First author.

ProcuraClaw: Context-Aware Procurement Search via Adaptive Hotel Profiling and Centralization Rate Prediction.

Under review at CIKM 2026. Co-first author.

RESEARCH PROJECTS

PilotBench: Safety-Critical Aviation Agents

Dec. 2025 – Mar. 2026

Core Contributor | Remote Research Collaboration | CESS, Beihang University

- Co-developed PilotBench from 708 real trajectories, 34-channel telemetry, and nine flight phases; contributed to evaluations of 41 models.
- Contributed to Pilot-Score design and analyses identifying a precision–controllability dichotomy and a dynamic-complexity gap in high-workload phases.

NeuroMark: Multimodal Clinical Reasoning

Sep. 2025 – Present

Research Contributor | Remote Research Collaboration | School of Data Science, Fudan University

- Led the initial project phase and evaluated 20 LLMs on 2,379 clinical questions and 16 VLMs across six brain-imaging tasks.
- Contributed to reliability analyses using safety compliance, effective accuracy, and difficulty discrimination; paired-model results showed no significant association between text and imaging performance ($\rho = 0.14$, $p = 0.76$).

ConcordCoder: Pre-generation Alignment for AI Coding

Mar. 2026 – Present

Online Research Intern | Waseda Institute for Advanced Study

- To prevent intent and compatibility errors before AI-generated patches, proposed ConcordCoder, which reconstructs code-grounded task context and asks developers to confirm constraints, risks, and non-goals before generation.
- Designed an inspectable **AlignmentRecord** and a controlled evaluation covering comprehension, review accuracy, evidence grounding, guard violations, and multi-round constraint retention.

Long-Horizon Memory Governance for LLM Agents (LHMG)

Dec. 2025 – May 2026

Project Lead | Remote Research Collaboration | School of Data Science, Fudan University

- Developed a prototype memory-governance layer with append-only revision records, active-state semantics, risk gating, and governed forgetting.
- In controlled diagnostics, observed a 13.3-point GPT-4o gain in cross-turn decision consistency over flat memory.

DDRG: Reasoning-Graph Verification and Repair

Apr. 2026 – Present

Research Collaborator | Li Lab

- Prototyped an inference-time reasoning-graph pipeline that samples candidate graphs, probes answer-sensitive claims, and gates local repairs conservatively.
- Evaluated Llama 3.1 70B on multiple reasoning benchmarks; case audits characterized parser, verifier, and repair-selection failures and informed conservative selection gates.

INDUSTRY EXPERIENCE

ProcuraClaw: Context-Aware Procurement Search

Nov. 2025 – Apr. 2026

AI Research Intern | H World Group

- Contributed to a deployed procurement-search system using LLM entity extraction, hybrid retrieval, profile-aware ranking, and Thompson Sampling.
- System results: MRR 0.95, NDCG@5 0.91 (175 queries); reported A/B tests showed 6.2%/9.8% first-hit CTR gains.

RESEARCH AND TECHNICAL SKILLS

AI and Programming: Multimodal learning, VLM/LLM evaluation, safety-critical AI, LLM agents, human–AI interaction, and machine learning; Python, PyTorch, C/C++, SQL, Java/JavaEE, Spring Boot, and Unity.

Research and Academic Communication: Benchmark and experiment design, model evaluation, ablation and error analysis, academic writing, and research presentations; Git, Linux, Docker, and $\text{\LaTeX}$.

Languages and Tests: Chinese (native); English (TOEFL iBT 90); Japanese (reading knowledge).